Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part II: transfer to MDEA, AMP, piperazine and DEA

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ABSTRACT

The shared water, carbon dioxide and inorganic-ion description established for monoethanolamine in Part I is held fixed while amine-specific parameters are estimated for methyldiethanolamine (MDEA), 2-amino-2-methyl-1-propanol (AMP), piperazine (PZ) and diethanolamine (DEA). The same equation of state accommodates their different reaction networks, but fitted pressures retain loading- and temperature-dependent errors, and PZ species residuals remain large relative to their assumed measurement scale. Held-out partial-pressure average absolute relative deviations (AARD) are 28.19 and 18.27 % for MDEA against Xiao and Ermatchkov, the latter mixing conditions inside and outside the dissociation law's measured temperature range; 42.85 % for AMP against Kundu; 21.94 % for PZ against Ermatchkov; and 79.54 and 44.17 % for DEA against Seo and Hong and Lal.

The data sets and fitting histories differ, so these percentages do not rank the amines. For AMP and DEA, local conditional parameter bands are much smaller than the unfitted-pressure discrepancies; MDEA bands overlap the errors, and PZ sensitivities limit the linear approximation. At least forty false liquid splits are detected for PZ across calibration and comparison states; a separate fit with Part I's stability penalty still has at least three, two of which the standard finite search missed. DEA reproduces fitted integral absorption heats within 0.98 % AARD but predicts the wrong sign of water–DEA excess enthalpy. The results establish the extent and limitations of amine-specific calibration under a frozen shared description, rather than uniform predictive accuracy across amine classes.

1. Introduction

A shared electrolyte description is useful only to the extent that it describes observations beyond those used to estimate it. Part I fitted an explicit-ion electrolyte perturbed-chain statistical associating fluid theory (ePC-SAFT) description of loaded aqueous monoethanolamine (MEA), in which every ion is a separate species and equilibrium, caloric and density properties follow from one Helmholtz energy [1]. That fit supplies the frozen water, carbon dioxide and inorganic-ion parameters used here. Its pressure average absolute relative deviations (AARDs) at unfitted MEA concentrations are 33.89 % at 15 wt% and 44.53 % at 45 wt%, despite closer agreement at the fitted concentration. Those errors motivate a test of the shared description across other amines; they do not supply a common accuracy standard for different experimental data.

Methyldiethanolamine (MDEA) is a tertiary amine whose reaction network contains bicarbonate formation without amine carbamate [2, Sect. 3]. 2-Amino-2-methyl-1-propanol (AMP) is a sterically hindered primary amine, diethanolamine (DEA) a secondary amine, and piperazine (PZ) a cyclic diamine with mono- and dicarbamate chemistry. AMP nevertheless forms measurable carbamate [3, Sects. 3.1–3.2, Fig. 5]. DEA forms carbamate alongside bicarbonate [4, Sects. 2–3, Fig. 8]; PZ also has protonated forms and protonated carbamate [5, Sects. 2–3, Table 5]. Measured species groups for these amines need not resolve each model species separately.

1.1. Thermodynamic descriptions of reactive MDEA, AMP, PZ and DEA

Activity-coefficient models combine reaction equilibria with nonideal liquid activities. Hessen et al. fit electrolyte non-random two-liquid (eNRTL) MDEA binary parameters to vapor–liquid equilibrium, excess enthalpy and freezing-point data, then fit reactive pressures and compare species [6, Sects. 4.1, 4.4–4.5]. Hartono et al.'s AMP eNRTL model includes carbamate, leaving species and absorption heat outside fitting [7, Sect. 5.2, Figs. 7–8]; their PZ model includes doubly protonated PZ and sequential pure, binary and reactive estimation [8, Sects. 2.1–2.3, 5.3–5.4, Tables 14–15]. Yi

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et al. instead omit AMP carbamate and doubly protonated PZ, and use pressure, heat-capacity constraints, parameter uncertainty and heat/species comparisons during optimization [9, Sects. 2.1–2.3, 3.1–3.2]. Their species comparisons therefore do not constitute blind validation.

Modified Pitzer and extended universal quasi-chemical (UNIQUAC) models provide other explicit-species descriptions. Silkenbäumer et al. combine single-amine AMP and MDEA parameters to predict blend solubility [10, pp. 3135–3140, Tables 5, 9]. Pérez-Salado Kamps et al. fit PZ solubility and predict MDEA/PZ mixtures; Ermatchkov et al. extend the PZ model with low-loading pressures [11, Modeling, pp. 2663–2664][12, Thermodynamic Modeling, pp. 1790–1793]. Böttinger et al. refit DEA pressure interactions and constrain carbamate chemistry with species observations [4, Sects. 5–7, Tables 7–8, Fig. 8]. Extended UNIQUAC modeling of AMP retains water, CO_2 and inorganic-ion inputs, fits AMP interactions in stages, and combines speciation, calorimetry and computational chemistry; the carbamate constant remains model-dependent [13, Sects. 4.1–4.4, 5.1–5.3]. Zhang and Chen fit MDEA–water eNRTL interactions to vapor–liquid equilibrium, excess enthalpy and heat capacity, then fit molecule–electrolyte interactions and $MDEAH^+$ standard-state properties to reactive pressures, absorption heats, heat capacities and nuclear magnetic resonance (NMR) species data [14, Sects. 3.2–3.3, Tables 12–15]. Their reactive-pressure fit spans 313–413 K and approximately 2–8 mol MDEA per kg water; source-specific average relative deviations distinguish fitted observations from pressure predictions for excluded sources, whose inconsistencies prevent agreement with all datasets [14, Sect. 3.3, Tables 14, 16]. Sadegh et al. estimate extended UNIQUAC MDEA parameters first from binary data and then from reactive data, using vapor pressures, freezing points, excess enthalpies, heat capacities and absorption heats; their ternary-pressure data cover 5–75 wt% MDEA and extend to 200 °C, and agreement is reported by data source and property [15, Sects. 7–9.1, Tables 5–7]. NMR species fractions at 20 and 40 °C in 23 wt% MDEA remain outside fitting and are assessed by absolute mole-fraction deviations [15, Sect. 9.2, Table 8]. Biget et al. apply staged extended UNIQUAC estimation to MEA, MDEA and AMP, fitting binary excess enthalpies and phase-equilibrium data before reactive solubilities and total pressures over a stated 40–150 °C and 15–60 mass% range, using a genetic algorithm followed by a quasi-Newton method [16, pp. 298–300, Fig. 1, Eq. 3]. Their property-specific AARDs and parameter confidence intervals describe the fits; the binary enthalpy and phase-envelope comparisons also use estimation data [16, pp. 300–302, Table 1, Figs. 2, 4, 6, Eqs. 4–5].

Published SAFT amine models distinguish parameter estimation from predictions to different extents. Nasrifar and Tafazzol organize MEA, DEA and MDEA around pure properties, binary interactions and subsequent acid-gas comparisons, using an ideal-activity treatment rather than explicit electrolyte activity coefficients [17, Sects. 2–6]. They lump ions with water for molecular PC-SAFT phase calculations; DEA species and acid-gas predictions follow the lower-order fits [17, Sect. 6, Figs. 13–17].

Reactive-association models describe reaction products as molecular association complexes. Rodriguez et al.'s SAFT for potentials of variable range (SAFT-VR) transfers functional-group association information for AMP, DEA and MDEA, while reactive ternary data determine remaining CO_2–amine parameters [18, Sects. 2.3, 3–5]. Its MDEA concentration prediction uses observations outside estimation, whereas the AMP and DEA curves include fitted conditions [18, Sects. 5.2–5.4, Figs. 10–12]. Transfer of molecular interactions thus coexists with reactive-data fitting.

Pereira and Vega's soft-SAFT treatment covers all four amines with implicit reactions, estimating association energies at fixed amine concentrations before testing other concentrations and predicting blends from single-amine interactions [19, Sects. 3.2–3.4, 4.1.1–4.1.2]. They compare alternative AMP association schemes, but do not resolve the complete PZ ionic network. Wang et al.'s Peng–Robinson cubic-plus-association (PR-CPA) treatment fits MDEA reaction-association and dispersion parameters to ternary data, then compares additional conditions and species inferred from bonded fractions [20, Sects. 3.2.2, 4.1.2, 4.2, Fig. 15]. These association models differ in both chemistry resolution and data use.

Group-contribution SAFT-γ Mie also transfers amine interactions while describing reactions implicitly. Perdomo et al. use lower-concentration PZ pressures to estimate cyclic-amine interactions and test a higher concentration [21, Sect. 4.2, Fig. 4]. Their DEA pressure and species predictions follow group-interaction estimation; MDEA pressures also enter estimation of tertiary-alkanolamine interactions [21, Sect. 4.3.3, Figs. 13–15]. These comparisons support transfer of group descriptions, but do not independently test explicit-ion parameters.

Uyan et al. inherit neutral and inorganic-ion parameters and derive $MDEAH^+$ inputs from molecular MDEA before predicting reactive pressures [2, Sects. 4.1–4.3, Tables 3–5]. Wangler et al. estimate MDEA ion–water interactions using osmotic data before predicting reactive ternary equilibria, and re-determine the ternary-pressure deviation of Uyan et al. on the same data [22, Sects. 6.1–6.2.1, Table 11]. Their absorption-heat predictions overstate temperature dependence at low loading [22, Sect. 6.3, Figs. 19–21]. Cleeton et al. inherit inorganic-ion inputs and estimate pure

and binary MDEA quantities, with larger concentrated-solution and low-loading pressure errors [23, Sects. 3.1–3.3.2, Figs. 7–8]. Raeispour Shirazi and Lotfollahi combine mean spherical approximation and Born terms with PC-SAFT and reactive-pressure fitting; their improvement also involves adjusted interactions [24, Eq. 12, Sect. 5.3, Tables 3–4]. Among the studies reviewed here, explicit-ion PC-SAFT has been applied only to MDEA; AMP, PZ and DEA have been described by activity-coefficient, implicit-association or group-contribution models.

1.2. Property constraints and phase stability

Species evidence adds distinct constraints: AMP peak splitting requires additional relations, PZ constants are inferred with activity assumptions, and DEA measurements combine molecular and protonated amine and identify byproducts [3, Sects. 2.5–3.2, Figs. 5, 7][5, Sects. 2–3, Table 5][4, Sects. 3, 5–7, Fig. 8]. Hartono's unfitted AMP heat comparison retains high-loading disagreement [7, Sect. 5.2, Fig. 8]. Thus species and caloric comparisons test more than pressure agreement; assigned residual scales and conditional parameter errors describe different uncertainties. Sadegh et al.'s objective assigns separate residual scales to equilibrium and caloric properties, reduces the weight of low-pressure data through a pressure offset, and leaves two absorption-heat datasets outside fitting [15, Sect. 7, Eq. 20, Sect. 9.1.2, Figs. 8–9]. Biget et al. instead minimize squared relative residuals in stages and estimate parameter confidence intervals from the objective Hessian and an estimated measurement variance; their multi-property fitting constraints differ from an unfitted-property test [16, pp. 298–302, Eqs. 3–5, Table 1].

The SAFT-γ Mie phase-changing cyclohexylamine examples and Tsanas et al.'s electroneutral liquid-phase searches for 2-(diethylamino)ethanol/3-(methylamino)propylamine (DEEA/MAPA) are analogs, not measured phase boundaries for MDEA, AMP, PZ or DEA [21, Sects. 4.4.1–4.4.2, Figs. 16–18][25, Sects. 2.5–2.7, 3.1, 3.3]. Tsanas et al. identify Gibbs–Duhem inconsistency in their eNRTL implementation. The present finite liquid-phase searches exclude solid equilibrium and cannot establish global stability.

1.3. Aim of this work

Here the water, CO_2 and inorganic-ion parameters from Part I remain fixed while only amine-specific quantities are fitted. The question is how far that description supports amine-specific calibration across these four structures when assessed against unfitted properties and searches for additional liquid phases. Each close precedent covers part of this combination: explicit-ion ePC-SAFT treats one tertiary amine, MDEA [2, 22, 23]; extended UNIQUAC retains shared water, CO_2 and inorganic-ion inputs for AMP [13]; and soft-SAFT and SAFT-γ Mie transfer interactions across amines but describe reactions implicitly [19, 21]. The four amines test the shared inorganic-carbon description, whose bicarbonate–water interaction lies on its upper bound in Part I [1], without an amine-carbamate pathway in MDEA and alongside different carbamate pathways in AMP, PZ and DEA. One example per class cannot isolate chemical class from differences in observation coverage, species representation or fitting freedom.

The present fits include amine-specific reactive observations, so agreement with those observations is calibration. In this paper, held-out observations never enter a fitting objective. That term does not imply that the data remained unseen throughout selection of the final fit. Previously accessed observations outside the fit and comparisons made after selecting the parameters are identified separately. Finite searches for additional phases provide numerical evidence about the calculated liquids, while experimental pressure and property comparisons measure agreement with the named observations. Neither form of evidence alone establishes independent validation of ion parameters or qualification for absorber design. The results follow the properties being tested, so fitted pressures, unfitted pressures, other thermodynamic properties and phase behavior each constrain the same transfer hypothesis.

2. Thermodynamic model

The water and CO_2 component descriptions, hydronium, hydroxide, bicarbonate and carbonate, all interactions among these six shared species, and the reaction laws of Part I, R1 (water autoprotolysis), R2 ($CO_2 + 2\,H_2O \rightleftharpoons HCO_3^- + H_3O^+$) and R3 ($HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$), including Part I's fitted R2 corrections, are held fixed [1]. Water segment, dispersion and association inputs were fitted independently to the IAPWS-95 reference equation of state for water; inorganic-ion segment sizes and energies are inherited from Uyan et al. [2, Table 4]. Table 1 distinguishes independent fits, literature values, parameters estimated jointly with MEA, and assumptions. Free interactions between an amine-specific ion and a shared species remain amine-specific parameters; freezing bicarbonate does not freeze every interaction involving bicarbonate.

The MDEA and AMP fits were first developed against an earlier MEA fit whose bicarbonate–water interaction was 0.727 rather than Part I's bound value +1 and whose R2 coefficients differed slightly (Supplementary Section S7); Section 4.3.2 tests that difference.

Table 1
Transferred quantities and their origins. All shared quantities remain fixed here; amine-specific interactions with shared species may be fitted.

Quantity	Origin in Part I	Implication here
Water segment, energy and association coefficients	Independent fit to IAPWS-95	Fixed; not fitted to any of these amines
CO_2 segment and energy coefficients	Gross and Sadowski, Table 2 [26]	Literature inputs; fixed
CO_2–water dispersion correction and slope	Independent fit to Kiepe pressures	Fixed
Inorganic-ion segment sizes and energies	Uyan et al., Table 4 [2]	Fixed; hydrogen-ion values used for hydronium
Inorganic Born sizes and remaining water–ion terms	Assumed inputs in Part I	Fixed; not independently measured in MEA
Bicarbonate–water interaction	MEA joint fit, upper bound +1	Fixed at bound; conditional SE excludes this direction
R2 level and enthalpy correction	MEA joint fit; interior with conditional errors	Fixed; source-law correction, not a new measurement
R1 and R3 water/carbonate laws	Literature laws as mapped in Part I	Fixed with source-reference conversion
Solvation and dielectric descriptions	Literature coefficients and assumptions in Part I	Water/CO2 inputs fixed; amine coefficients specified separately
Fitted reactive parameters (MDEA / AMP / PZ / DEA)	Amine-specific fits: 9 / 9 / 12 / 4	Not totals of earlier pure and binary fitting

2.1. Residual Helmholtz energy

The reduced residual Helmholtz energy $a^{\mathrm{res}} = A^{\mathrm{res}}/(nRT)$ combines hard-chain repulsion, dispersion, association and electrostatic contributions [26, 27, 28, 29, 30]:

$$a^{\mathrm{res}} = a^{\mathrm{hc}} + a^{\mathrm{disp}} + a^{\mathrm{assoc}} + a^{\mathrm{DH}} + a^{\mathrm{Born}}. \tag{1}$$

Here n is the total amount of true species, $x_i = n_i/n$, $\rho = n/V$, and $\rho_N = N_{\mathrm{A}}\rho$ is number density. Neutral species are chains of m_i^{seg} segments, with segment diameter σ_i and dispersion energy ϵ_i; the amine-specific charged-chain exceptions are specified below. Pressure and chemical potentials follow from volume and composition derivatives of this potential, including the density, temperature and composition dependence of the relative permittivity.

The hard-chain reference uses the hard-sphere mixture contact function g_{ij}^{hs} and packing fraction $\eta = \zeta_3$ [26, Eqs. A.4–A.8]:

$$a^{\mathrm{hc}} = \bar{m}\, a^{\mathrm{hs}} - \sum_i x_i \left(m_i^{\mathrm{seg}} - 1\right) \ln g_{ii}^{\mathrm{hs}}, \qquad \bar{m} = \sum_i x_i m_i^{\mathrm{seg}}, \tag{2}$$

$$a^{\mathrm{hs}} = \frac{1}{\zeta_0}\left[\frac{3\zeta_1\zeta_2}{1-\zeta_3} + \frac{\zeta_2^3}{\zeta_3\left(1-\zeta_3\right)^2} + \left(\frac{\zeta_2^3}{\zeta_3^2} - \zeta_0\right)\ln\left(1-\zeta_3\right)\right], \tag{3}$$

$$g_{ij}^{\mathrm{hs}} = \frac{1}{1-\zeta_3} + \frac{d_i d_j}{d_i + d_j}\frac{3\zeta_2}{\left(1-\zeta_3\right)^2} + \left(\frac{d_i d_j}{d_i + d_j}\right)^2 \frac{2\zeta_2^2}{\left(1-\zeta_3\right)^3}, \qquad \zeta_n = \frac{\pi}{6}\rho_N \sum_i x_i m_i^{\mathrm{seg}} d_i^n, \quad n = 0, \ldots, 3. \tag{4}$$

2.2. Segment diameters and dispersion interactions

Neutral species and the charged-chain cations $MDEAH^+$ and $DEAH^+$ use the temperature-dependent packing diameter of Gross and Sadowski [26, Eq. A.9]:

$$d_i = \sigma_i \left[1 - 0.12 \exp\left(-\frac{3\epsilon_i}{k_B T}\right)\right]. \tag{5}$$

Water retains the exponential temperature dependence tabulated by Pabsch et al. [31, Table 2], with its constant term fitted to IAPWS-95 in Part I [1]:

$$\sigma_w(T) = 2.7804326 + 10.11 \exp(-0.01775T) - 1.417 \exp(-0.01146T), \tag{6}$$

where σ_w is in Å and T in K. All other ions use packing diameters $0.88\sigma_i$. Debye–Hückel diameters equal σ_i for every ion; Born diameters are separate parameters. The MDEA cation retains the neutral segment count, whereas the DEA cation retains the neutral segment count, diameter and dispersion energy. AMP and PZ ions and the AMP and DEA carbamates use single spheres; PZ cation and carbamate ties are given below.

The pair rules are $\sigma_{ij} = (\sigma_i + \sigma_j)/2$ and

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j}\,\left(1 - k_{ij}\right). \tag{7}$$

Cation–anion and ion–solvent dispersion are retained. Like-charge dispersion is excluded as in Held et al. [29, Sect. 2.3], except that the charged-chain $MDEAH^+$ and $DEAH^+$ representations retain their self-dispersion, with $\epsilon_{ii} = \epsilon_i$. These charged-chain exceptions are choices of the present model. The dispersion perturbation is [26, Eqs. A.10–A.19, Table 1]

$$a^{\mathrm{disp}} = -2\pi\rho_N I_1 \sum_i \sum_j x_i x_j m_i^{\mathrm{seg}} m_j^{\mathrm{seg}} \frac{\epsilon_{ij}}{k_B T} \sigma_{ij}^3 - \pi\rho_N \bar{m}\, C_1 I_2 \sum_i \sum_j x_i x_j m_i^{\mathrm{seg}} m_j^{\mathrm{seg}} \left(\frac{\epsilon_{ij}}{k_B T}\right)^2 \sigma_{ij}^3, \tag{8}$$

$$C_1 = \left[1 + \bar{m}\frac{8\eta - 2\eta^2}{(1-\eta)^4} + (1-\bar{m})\,\frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{\left[(1-\eta)(2-\eta)\right]^2}\right]^{-1}, \tag{9}$$

$$I_1 = \sum_{k=0}^{6} a_k(\bar{m})\,\eta^k, \qquad I_2 = \sum_{k=0}^{6} b_k(\bar{m})\,\eta^k, \qquad a_k(\bar{m}) = a_{0k} + \frac{\bar{m}-1}{\bar{m}} a_{1k} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}} a_{2k}. \tag{10}$$

The b_k coefficients have the same chain-length form as a_k; their universal coefficients are those of Gross and Sadowski, Table 1.

2.3. Association scheme

Association uses the site fractions X_A and bond strengths Δ^{AB} of first-order perturbation theory [27, Sect. 2][32, Eqs. 4–8]:

$$a^{\mathrm{assoc}} = \sum_i x_i \sum_{A\in i} \left(\ln X_A - \frac{X_A}{2} + \frac{1}{2}\right), \qquad X_A = \left[1 + \rho_N \sum_j x_j \sum_{B\in j} X_B \Delta^{AB}\right]^{-1}, \tag{11}$$

$$\Delta^{AB} = g_{ij}^{\mathrm{hs}}\, \sigma_{ij}^3\, \kappa^{AB} \left[\exp\left(\frac{\epsilon^{AB}}{k_B T}\right) - 1\right]. \tag{12}$$

The sums include each site's multiplicity, and only the declared donor–acceptor pairs bond. Association energies ϵ^{AB} are distinct from dispersion energies ϵ_i; κ^{AB} is dimensionless. Unlike amine–water association uses

$$\epsilon_{ij}^{AB} = \tfrac{1}{2}\left(\epsilon_i^{AB} + \epsilon_j^{AB}\right), \qquad \kappa_{ij}^{AB} = \sqrt{\kappa_i^{AB}\kappa_j^{AB}}\left(\frac{\sqrt{\sigma_i\sigma_j}}{\sigma_{ij}}\right)^3. \tag{13}$$

Water, AMP and PZ have one donor and one acceptor site; MDEA and DEA have two of each. These are effective association descriptions, not counts of reactive species. Water–CO_2 induced association has one site of each role on CO_2, with no self-association, energy $\epsilon_w^{AB}/2$ and volume κ_w^{AB} [33, Sect. 2.4.2, Table 1][31, Modeling]. There is no amine–CO_2 association bond. The MDEA and DEA cations carry the neutral site multiplicities and association volumes but zero self-association energy; their cross-association with water still uses the unlike rule. AMP and PZ ions, carbamates and the neutral PZ zwitterion HPZCOO carry no association sites. The amine interaction tables in Section 2.8 give self- and cross-association energies and volumes in the $\sigma_{ij}^3\kappa^{AB}$ convention.

2.4. Debye–Hückel and Born contributions

The finite-size Debye–Hückel contribution uses the ion set $\mathcal{I}$, charge number z_i and inverse screening length κ_D [28, Eqs. 5, 13–14]:

$$a^{DH} = -\frac{\kappa_D e^2}{12\pi\varepsilon_0\varepsilon_{bulk}k_B T}\sum_{i\in\mathcal{I}} x_i z_i^2 \chi_i, \qquad \kappa_D^2 = \frac{e^2\rho_N}{\varepsilon_0\varepsilon_{bulk}k_B T}\sum_{i\in\mathcal{I}} x_i z_i^2, \tag{14}$$

$$\chi_i = \frac{3}{\left(\kappa_D d_i^{DH}\right)^3}\left[\frac{3}{2} + \ln\left(1+\kappa_D d_i^{DH}\right) - 2\left(1+\kappa_D d_i^{DH}\right) + \frac{1}{2}\left(1+\kappa_D d_i^{DH}\right)^2\right]. \tag{15}$$

Diameters and number density use consistent SI units; the zero-screening limit is $\chi_i = 1$.

The solvation-shell and dielectric-saturation form uses Figiel et al.'s model with its corrected Eq. 7 [30, Eqs. 5–10][34]:

$$a^{Born} = -\frac{e^2}{4\pi\varepsilon_0 k_B T}\sum_{i\in\mathcal{I}} x_i z_i^2\left[\frac{1-\varepsilon_{bulk}^{-1}}{D_i} + c_{dielectric}\left(1-\varepsilon_{ion}^{-1}\right)\left(\frac{1}{d_i^{Born}} - \frac{1}{D_i}\right)\right], \tag{16}$$

$$D_i = d_i^{Born}\left[1 + c_{shell}\frac{f_{mix}-1}{|z_i|}\right], \qquad f_{mix} = \frac{\sum_k x_k f_k}{\sum_k x_k}. \tag{17}$$

Here d_i^{Born} is the ion diameter, D_i the outer shell diameter, and $\varepsilon_{ion} = 8$ the assumed local relative permittivity. Both switches c_{shell} and $c_{dielectric}$ equal unity. The water solvation factor is $f_w = 1.5$ [30, Table 2]; all other species have $f_k = 1$. The average here includes every species, giving $f_{mix} = 1 + 0.5x_w$, rather than restricting the average to ions and solvents. The tabulated amine-ion Born diameters are fixed or fitted effective quantities, not independent solvation measurements.

2.5. Relative permittivity

The Schreckenberg mixture form reported by Yang et al. supplies the bulk relative permittivity in both electrostatic contributions [35, Sect. 2.1, Eqs. 1a, 2a–2b]:

$$\varepsilon_{bulk} = 1 + \frac{n_s}{V}d_{V,mix}\left(\frac{d_{T,mix}}{T} - 1\right), \qquad d_{q,mix} = \frac{\sum_{i\in\mathcal{S}} n_i d_{q,i}}{n_s}, \quad q \in \{V, T\}. \tag{18}$$

The neutral set $\mathcal{S}$ contains water, molecular CO_2 and molecular amine, plus HPZCOO for PZ; n_s is their total amount. The dielectric coefficients $d_{V,i}$ and $d_{T,i}$ have units of volume per mole and kelvin, with n_s/V in matching units. Ions affect volume and neutral composition but are excluded from the mixing averages. The law approaches unity at zero density. MDEA, AMP and DEA coefficients use Hsieh et al.'s unloaded measurements at model liquid densities [36, Tables 1–2, 5]; PZ and HPZCOO use the MEA coefficients fitted in Part I as an assumption [1]. Extension beyond the dielectric measurements and to loaded solutions is a model assumption.

2.6. Reactions and phase equilibrium

Each liquid contains the shared species and its molecular amine and protonated forms. AMP and DEA also contain carbamate; PZ contains singly and doubly protonated PZ, mono- and dicarbamate, and the neutral zwitterion HPZCOO.

For MDEA, AMP and DEA, amine dissociation is $AH^+ + H_2O \rightleftharpoons A + H_3O^+$; AMP and DEA carbamate hydrolysis is $ACOO^- + H_2O \rightleftharpoons A + HCO_3^-$. PZ has the five reaction directions

$$\begin{aligned}
&\text{P1}: \quad PZH^+ + H_2O \rightleftharpoons PZ + H_3O^+,\\
&\text{P2}: \quad PZH_2^{2+} + H_2O \rightleftharpoons PZH^+ + H_3O^+,\\
&\text{P3}: \quad PZCOO^- + H_2O \rightleftharpoons PZ + HCO_3^-,\\
&\text{P4}: \quad HPZCOO + H_2O \rightleftharpoons PZCOO^- + H_3O^+,\\
&\text{P5}: \quad PZ(COO)_2^{2-} + H_2O \rightleftharpoons PZCOO^- + HCO_3^-.
\end{aligned} \tag{19}$$

The fitted ion diameters and reaction-law corrections are effective model parameters, not independently measured molecular sizes or equilibrium constants.

Charged species remain liquid-only. The vapor contains water, CO_2 and molecular amine, and for PZ also HPZCOO. Allowing this neutral zwitterion in the vapor resolves otherwise undecided phase searches; with the PZ parameters held fixed, the relative change in fitted partial pressures is at most 1.2×10^{-8}. This numerical change is negligible for those pressures, but the vapor composition space matters to the phase search. PZ has no solid-phase description.

Water uses a pure-liquid reference; every solute uses infinite dilution in water. The amine laws and shared R2 retain their source molality standard state; R1 and R3 retain the source mole-fraction basis. Water activity is explicit, and all reactions use the hydronium reference. For a reaction written with stoichiometric coefficients ν_i positive for products, equilibrium requires

$$\sum_i \nu_i \mu_i = 0, \qquad \ln K(T) = a + b/T + c\ln(T/T_0) + dT + \cdots . \tag{20}$$

Here $T_0 = T_{\mathrm{ref}} = 313.15$ K, the reference temperature also used in Eq. (23). Chemical potentials of each vapor-allowed species are equal between liquid and vapor, and element balances and electroneutrality constrain each reacting liquid. For $Z = P/(\rho RT)$ and $\ln\phi_i = \mu_i^{\mathrm{res}}/(RT) - \ln Z$, the dimensionless activities are

$$a_w = x_w \frac{\phi_w(T,P,\boldsymbol{x})P}{\phi_w^{\mathrm{pure}}(T,P^\circ)P^\circ}, \qquad a_i = \frac{x_i}{M_w m^\circ}\frac{\phi_i(T,P,\boldsymbol{x})P}{\phi_i^{\infty,w}(T,P^\circ)P^\circ} \quad (i \neq w). \tag{21}$$

Here M_w is water's molar mass in kg/mol, $m^\circ = 1$ mol/kg and P° is the source reference pressure; $\phi_i^{\infty,w}$ is evaluated along a charge-neutral dilution path. Only charge-neutral reaction combinations require the ionic reference. Thus $a_i = (m_i/m^\circ)\gamma_i^{(m)}$, where $m_i = x_i/(M_w x_w)$ and $\gamma_i^{(m)} = x_w\phi_i P/(\phi_i^{\infty,w}P^\circ)$. In terms of these activities, the reaction condition of Eq. (20) becomes $\sum_i \nu_{ri}\ln a_i = \ln K_r$.

The source constants are converted once according to

$$\ln K_{m,r} = \ln K_{x,r} - \Big(\sum_{i\neq \mathrm{H_2O}} \nu_{ri}\Big)\ln\Big(M_{\mathrm{H_2O}} m^\circ\Big). \tag{22}$$

The shared water and inorganic-carbon laws use Austgen et al.'s correlation form [37, Table V]. R1–R3 retain their system-pressure reference; amine laws use their source standard pressure. A fitted constant-enthalpy correction δh_r leaves $K_r(T_{\mathrm{ref}})$ unchanged:

$$\delta \ln K_r(T) = -\frac{\delta h_r}{R}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right). \tag{23}$$

For AMP and DEA, the fitted hydrolysis level is the constant coefficient a of the hydrolysis law in Eq. (20); because Eq. (23) also shifts a by $\delta h_r/(RT_{\mathrm{ref}})$, the tabulated a differs from the fitted level.

Table 2
Reaction-law measurement intervals and use. A calculation inside a calculation interval can still extrapolate the measurements. Rounded source limits are distinguished from actual tabulated temperatures.

Law	Measured interval / K	Calculation interval / K	Use and limitation
MDEA dissociation	278–368	278–395	393.15 K pressure and 388.71 K heat extrapolate measurements
AMP dissociation	to 353.22	to 353.22	Tong 373–393 K rows not calculated (outside calculation interval)
AMP hydrolysis	source estimates: 313–373	includes 303–353	Not direct law measurements; slope extrapolated below 313 K; level and slope fitted
PZ protonation	293.14–353.42	includes 283.15–395.0	283.15 K and above 353.42 K extrapolate
PZ carbamate	283.15–333.15 (rounded 283–333)	includes 353.15–395.0	Above 333.15 K extrapolates
DEA protonation	to approximately 323	to 393.15	23/30 fitted Lawson rows above measured range
Shared R2	not established here	293.15–393.15	Seven Lawson rows at 394.26 K not calculated (outside calculation interval); PZ 283.15 K extrapolates below lower limit; Kamps 395.0 K above upper limit

A state can lie inside a reaction law's calculation interval, the temperatures over which this work evaluates the law, yet outside the temperatures at which that law was measured. Table 2 lists both intervals for each law. The MDEA dissociation coefficients are those of Kamps and Maurer [38], measured over 278–368 K, although the calculation interval extends to 395 K. The AMP hydrolysis starting slope is inferred from Xu et al.'s model-derived carbamate constants [39, Table 2, Eqs. 7–11]; it is not an independently measured reaction-temperature law. The fixed AMP dissociation law [40] ends at 353.22 K. Its carbamate-hydrolysis temperature dependence is fitted: the fitted b is -1944.8 K, compared with the starting source slope -2546.2 K, while the enthalpy correction reaches its -5000 J/mol limit. This fitted hydrolysis slope is distinct from the fixed dissociation law and does not extend that law's admissible interval. PZ carbamate source measurements span 283.15–333.15 K, and its protonation measurements approximately 293–353 K. DEA's hydrolysis starting slope is the inverse of Aroua et al.'s formation law [41, Eq. 14]; the level and enthalpy correction are fitted. DEA uses the fixed Bower protonation law [42] above the approximately 323 K upper measurement limit, including 23 of its 30 fitted pressures. A fitted offset does not turn an extrapolated temperature dependence into a measured one.

2.7. Enthalpy and heat capacity

Total enthalpy follows the same potential as equilibrium:

$$H = PV - RT^2 \left(\frac{\partial (A/RT)}{\partial T} \right)_{V,\boldsymbol{n}}, \qquad A = A^{\mathrm{ig}} + A^{\mathrm{res}}. \tag{24}$$

Neutral ideal-gas functions provide the caloric reference; reaction laws complete the ionic functions with $\Delta_r h^\circ = RT^2\, d \ln K_r/dT$ and the derivatives of the infinite-dilution corrections. Reaction heats therefore enter once through the standard-state functions. The shared CO_2 ideal-gas function follows NIST-JANAF [43]; water uses the ideal-gas part of IAPWS-95 [44, Eq. 5, Table 1]. MDEA uses the Neumann ideal-gas correlation and DEA uses a linear ideal-gas heat-capacity fit obtained from liquid calorimetry after subtracting the residual contribution; no independent absorption-heat term is added. Equilibrium isobaric heat capacity is $C_P = (dH_{\mathrm{eq}}/dT)_{P,\boldsymbol{n}_{\mathrm{feed}}}$, including the change in speciation. The molar-feed quantity is C_P/n_{feed} and the mass-specific quantity is $c_P = C_P/m_{\mathrm{feed}}$, with the complete analytical-feed mass including added CO_2.

2.8. Amine-specific inputs

MDEA retains the four-site neutral parameters of Pahlavanzadeh and Fakouri Baygi [45, Sect. 3.1, Table 2], whose pure-component fit uses saturation-property correlations, and its fixed dissociation law. Its association volume is adjusted to reproduce that neutral source model with the σ_{ij}^3 association-strength convention used here, while the segment count, diameter, dispersion and association energy remain fixed. The amine–water binary observations are those of Kim et al. [46, Table 8]; the neutral interaction is released again in the reactive fit. Its ideal-gas heat capacity is a cubic approximation to Neumann et al.'s correlation [47, Eq. 5, Table 8]; its dielectric coefficients are fitted to Hsieh et al. [36, Tables 2, 5]. Its final fit releases the neutral–water interaction and linear temperature slope, cation interactions with water, CO_2 and bicarbonate, a reciprocal-temperature slope for cation–water, and cation segment diameter, dispersion energy and Born diameter.

The neutral–water law is $k(T) = k_0 + k_1(T - 313.15\,\mathrm{K})$, with $k_0 = -0.2063373482$ and $k_1 = 0.0014750401\ \mathrm{K}^{-1}$. The cation–water law is $k(T) = k_0 + b_{ij}(1/T - 1/313.15\,\mathrm{K})$ with $k_0 = -0.03323620783$ and $b_{ij} = -27.10492507$ K. The nine fitted parameters do not include a fitted dissociation constant.

Table 3
MDEA amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
MDEA	m^{seg}	3.675	1
MDEA	σ	3.563	Å
MDEA	ϵ/k_B	228.711	K
MDEA	f	1	1
MDEA	d_V	0.86306697	L/mol
MDEA	d_T	1121.7735	K
$MDEAH^+$	m^{seg}	3.675	1
$MDEAH^+$	σ	3.6494074	Å
$MDEAH^+$	ϵ/k_B	623.04303	K
$MDEAH^+$	f	1	1
$MDEAH^+$	d^{DH}	3.6494074	Å
$MDEAH^+$	d^{Born}	7	Å

Table 4
MDEA nonzero constant pair coefficients and association parameters. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–$MDEAH^+$	k_{ij}	0.2481889
water–$MDEAH^+$	k_{ij}	−0.0332362
water–$MDEAH^+$	b_{ij} / K	−27.1049251
HCO_3^-–$MDEAH^+$	k_{ij}	1.0000000
MDEA–MDEA	ϵ^{AB}/k_B / K	2046.6240000
MDEA–MDEA	κ^{AB}	0.1180069
$MDEAH^+$–$MDEAH^+$	ϵ^{AB}/k_B / K	0.0000000
$MDEAH^+$–$MDEAH^+$	κ^{AB}	0.1180069

AMP's neutral description starts from Ayad et al.'s two-site parameters [48, Sects. 3.1, 4.5, Table 5] and is refitted to Klepáčová et al. and Belabbaci et al. vapor pressures and Ayad et al. densities [49, Table 10][50, Table 2][48, Table 2]. The binary fit uses Hartono et al. pressure and vapor-composition observations [51, Table 3], while the separate dielectric fit uses Hsieh et al. [36, Tables 2, 5]. AMP uses this neutral association description with explicit carbamate chemistry and a fixed Hamborg dissociation law [40]. The two temperature-dependent AMP–water and $AMPH^+$–water interactions have the reciprocal-temperature form above, with slopes −55.96106619 and −274.1548086 K,

respectively. Its final nine fitted parameters include those four interaction quantities, cation–CO_2 interaction, cation and carbamate diameters, hydrolysis level and hydrolysis enthalpy correction.

Table 5
AMP amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
AMP	ϵ/k_B	237.05992	K
AMP	m^{seg}	4.0761318	1
AMP	σ	3.1870454	Å
AMP	f	1	1
AMP	d_V	0.32290165	L/mol
AMP	d_T	2055.8807	K
$AMPH^+$	d^{Born}	5	Å
$AMPH^+$	d^{DH}	5.4152589	Å
$AMPH^+$	ϵ/k_B	237.05992	K
$AMPH^+$	d^{pack}	4.7654278	Å
$AMPH^+$	m^{seg}	1	1
$AMPH^+$	σ	5.4152589	Å
$AMPH^+$	f	1	1
carbamate	d^{Born}	5	Å
carbamate	d^{DH}	5.3199952	Å
carbamate	ϵ/k_B	207.21695	K
carbamate	d^{pack}	4.6815957	Å
carbamate	m^{seg}	1	1
carbamate	σ	5.3199952	Å
carbamate	f	1	1

Table 6
AMP nonzero constant pair coefficients and association parameters. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–$AMPH^+$	k_{ij}	0.7549873
AMP–water	k_{ij}	−0.1307372
AMP–water	b_{ij} / K	−55.9610662
carbamate–water	k_{ij}	−0.0036580
$AMPH^+$–water	k_{ij}	−0.3136388
$AMPH^+$–water	b_{ij} / K	−274.1548086
AMP–AMP	ϵ^{AB}/k_B / K	2385.1893365
AMP–AMP	κ^{AB}	0.0366211

PZ uses Razavi et al.'s effective two-site neutral association description [52, Sect. 3.1, Table 1, Eq. 7]; the pure fit holds segment count 2.6519 and diameter 3.7136 Å fixed while fitting its three energy/association quantities to Steele pressures [53, Table 7]. Pure-liquid density and the liquid extrapolation below the melting region remain unvalidated. The neutral PZ–water interaction has the linear-temperature form anchored at 353.15 K, with $k_0 = -0.1976454016$ and $k_1 = 0.0011250148\ \mathrm{K}^{-1}$; its two fitted parameters were fitted separately to binary pressure and vapor composition, then fixed during the reactive fit [51, Table 6]. The reactive fit with twelve fitted parameters ties the singly and doubly protonated cation sizes and energies; mono- and dicarbamate share size, energy and water interaction. Four charged PZ species have fixed 5 Å Born diameters. The neutral HPZCOO zwitterion has independent size and energy and no association sites. The remaining fitted quantities are the cation–water and cation–bicarbonate interactions and three additive offsets to $\ln K$ of P3–P5.

Table 7
PZ amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
PZ	m^{seg}	2.6519	1
PZ	σ	3.7136	Å
PZ	ϵ/k_B	313.20598	K
PZ	f	1	1
PZ	d_V	0.73098659	L/mol
PZ	d_T	1019.0803	K
PZH^+	m^{seg}	1	1
PZH^+	σ	4.2856111	Å
PZH^+	ϵ/k_B	102.72806	K
PZH^+	d^{Born}	5	Å
PZH^+	d^{pack}	3.7713378	Å
PZH^+	d^{DH}	4.2856111	Å
PZH^+	f	1	1
PZH_2^{2+}	m^{seg}	1	1
PZH_2^{2+}	σ	4.2856111	Å
PZH_2^{2+}	ϵ/k_B	102.72806	K
PZH_2^{2+}	d^{Born}	5	Å
PZH_2^{2+}	d^{pack}	3.7713378	Å
PZH_2^{2+}	d^{DH}	4.2856111	Å
PZH_2^{2+}	f	1	1
PZ carbamate	m^{seg}	1	1
PZ carbamate	σ	4.86247	Å
PZ carbamate	ϵ/k_B	717.58778	K
PZ carbamate	d^{Born}	5	Å
PZ carbamate	d^{pack}	4.2789736	Å
PZ carbamate	d^{DH}	4.86247	Å
PZ carbamate	f	1	1
PZ dicarbamate	m^{seg}	1	1
PZ dicarbamate	σ	4.86247	Å
PZ dicarbamate	ϵ/k_B	717.58778	K
PZ dicarbamate	d^{Born}	5	Å
PZ dicarbamate	d^{pack}	4.2789736	Å
PZ dicarbamate	d^{DH}	4.86247	Å
PZ dicarbamate	f	1	1
HPZCOO	m^{seg}	2.6519	1
HPZCOO	σ	3.8142737	Å
HPZCOO	ϵ/k_B	975.42757	K
HPZCOO	f	1	1
HPZCOO	d_V	0.73098659	L/mol
HPZCOO	d_T	1019.0803	K

Table 8
PZ nonzero constant pair coefficients and association parameters. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
water–PZH^+	k_{ij}	−1.0000000
water–PZH_2^{2+}	k_{ij}	−1.0000000

Pair	Quantity	Value
water–PZ carbamate	k_{ij}	0.0740702
water–PZ dicarbamate	k_{ij}	0.0740702
HCO_3^-–PZH^+	k_{ij}	−0.0210355
HCO_3^-–PZH_2^{2+}	k_{ij}	−0.0210355
PZ–PZ	ϵ^{AB}/k_B / K	1469.8200923
PZ–PZ	κ^{AB}	0.0272299

DEA neutral pure parameters use Nasrifar and Tafazzol's four-site scheme, with two donor and two acceptor sites [17, Table 2, Eq. 34]. Their association strength uses $\sigma^3\kappa$, consistent with the present implementation. The fitted neutral DEA–water interaction is temperature independent, $k = -0.0636$, and the dielectric coefficients were fitted separately to Hsieh et al.'s unloaded permittivities [36, Tables 1, 5] before reactive estimation. The binary interaction uses Horstmann et al.'s water–DEA pressures [54, Table 4]. DEA's ideal-gas heat capacity is a linear fit to Maham et al.'s liquid heat capacities minus the model residual contribution [55, Table 1]. The reported reactive fit represents $DEAH^+$ as a charged chain carrying DEA's segment count, diameter, dispersion energy and four sites, with zero self-association energy. The carbamate retains a single-sphere representation. Only the two ion–water interactions and hydrolysis level and enthalpy correction are fitted in the final objective with four fitted parameters. The alternative single-sphere cation fit is retained only as a comparison.

Table 9
DEA amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
DEA	ϵ/k_B	269	K
DEA	m^{seg}	5.2048	1
DEA	σ	2.9477	Å
DEA	f	1	1
DEA	d_V	0.75748303	L/mol
DEA	d_T	1155.7835	K
$DEAH^+$	d^{Born}	5	Å
$DEAH^+$	d^{DH}	2.9477	Å
$DEAH^+$	ϵ/k_B	269	K
$DEAH^+$	m^{seg}	5.2048	1
$DEAH^+$	σ	2.9477	Å
$DEAH^+$	f	1	1
carbamate	d^{Born}	5	Å
carbamate	d^{DH}	6.1672093	Å
carbamate	ϵ/k_B	183.00063	K
carbamate	d^{pack}	5.4271442	Å
carbamate	m^{seg}	1	1
carbamate	σ	6.1672093	Å
carbamate	f	1	1

Table 10
DEA nonzero constant pair coefficients and association parameters. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
DEA–water	k_{ij}	−0.0636335
carbamate–water	k_{ij}	0.0057148

Pair	Quantity	Value
$DEAH^+$–water	k_{ij}	−0.2151637
DEA–DEA	ϵ^{AB}/k_B / K	1517.6000000
DEA–DEA	κ^{AB}	0.1014000
$DEAH^+$–$DEAH^+$	ϵ^{AB}/k_B / K	0.0000000
$DEAH^+$–$DEAH^+$	κ^{AB}	0.1014000

3. Data and parameter estimation

Loading is $\alpha = n_{CO_2,added}/n_{amine,total}$, per mole of amine molecules, including for PZ. Unloaded mass fraction is $w_A = m_A/(m_A + m_w)$; solute-free stock compositions are retained for pressure calculations. True liquid mole fractions use the total number of molecular and ionic species, whereas PZ NMR group fractions use total PZ as denominator. The protonated-carbamate group and dicarbamate group are therefore not directly comparable to DEA true mole fractions. Total pressure and CO_2 partial pressure are separate observables.

Table 11
Observation roles and prior access. Held out means never in any fitting objective. Comparison-only means calculated after parameter selection without entering that selection. Prior scoring is stated even when no held-out value determined the objective.

Amine / role	Observations	Scored before final fit chosen?
MDEA fitted	278 entries: 67 Kuranov; 16 Jakobsen; 40 Carson; 4 Merkley; 33 heat capacities; 33 loaded and 24 unloaded densities; 61 Kim binary pressures	—
MDEA held-out	20 Xiao and 88 Ermatchkov partial pressures	Yes; previously accessed in model development
MDEA exclusions	Kuranov source rows 28 and 72; seven states above the 393.15 K R2 limit	Not counted as available predictions
AMP fitted	156 entries: 47 Hartono loaded total pressures; 52 Hartono binary entries; 57 Ciftja fractions	—
AMP held-out	51 Kundu partial pressures	Yes; scored during the sequence preceding final diameter choice
AMP held-out comparison	17 Tong total pressures scored; 23 not calculated	Earlier fits had been compared; no blind test claimed
PZ fitted	294 entries: 62 Hilliard pressures; 232 NMR group observations at 121 states	—
PZ held-out	52 Ermatchkov partial pressures	Yes; unconstrained refit scored before penalty alternative and final choice
PZ comparison-only	94 Kamps total pressures; precipitation row excluded	No; scored once on the fixed twelve-parameter fit
DEA fitted	80 entries: 30 Lawson pressures at 25 wt%; 50 Carson heats	—
DEA held-out	16 Seo and Hong; 44 Lal partial pressures	Yes; first selected start, then numerically tied second start
DEA comparison-only	Lawson Table VIII; Cai binary; Böttinger species; Horstmann excess heat; Weiland densities	No; added after fit selection

MDEA uses 278 objective rows: Kuranov et al. total pressures [56], Jakobsen et al. protonation fractions [57], Carson et al. [58] and Merkley et al. [59] heats, Weiland et al. heat capacities [60] and loaded densities [61], Sobrino et al. unloaded densities [62] and Kim et al. binary pressures [46]. Its Xiao [63] and Ermatchkov [64] pressure sets had been accessed during earlier development and are held-out assessments rather than new blind tests. AMP uses 47 loaded total pressures [65], 52 binary observations [51] and 57 carbamate fractions [3]; its Kundu set [66] was scored before the final diameter restriction was chosen. PZ uses 62 Hilliard partial pressures [67] and 232 NMR group observations from 121 source states [5]. Its Ermatchkov pressure set was scored on the unconstrained refit before the penalty alternative was examined and the reported fit was finalized. DEA uses 30 Lawson partial pressures [68] at 25

wt% and 50 Carson integral heats at 298.15 K; its held-out sets were first scored on one start and then on a numerically tied second start. Thus held-out here has the same objective-exclusion meaning for every amine, with prior access reported explicitly.

Carrier gases are omitted from the bubble calculations: the model compares $y_{\mathrm{CO_2}} P$ with measured partial pressure rather than imposing a nitrogen- or methane-containing experimental gas. This affects the MDEA and PZ nitrogen qualifications and the unreported methane pressure in the Lawson DEA cell. Lal's DEA stock, 2.0 kmol/m^3, is represented by mass fraction 0.2058 using unloaded-solution densities. Böttinger DEA comparisons [4] use 2.5 MPa because source cell pressures span 5–25 bar without individual values. PZ NMR solvent masses reported for heavy water are treated as water, leaving isotope effects unquantified. These assumptions are part of the compared states, not corrections inferred from the pressure errors.

3.1. Caloric and density calculations

For an initial stock i and final loaded equilibrium liquid f, the integral molar absorption enthalpy is

$$\Delta h_{\mathrm{abs,int}} = \frac{H_f - H_i - \Delta n_{\mathrm{CO_2}} h^{\mathrm{feed}}_{\mathrm{CO_2}}}{\Delta n_{\mathrm{CO_2}}}. \tag{25}$$

The feed and both liquids are evaluated at the comparison temperature and pressure; $\Delta n_{\mathrm{CO_2}}$ is the absorbed amount; $h^{\mathrm{feed}}_{\mathrm{CO_2}}$ is the molar enthalpy of the CO_2 feed; negative heat denotes exothermic absorption. For these negative heats, the signed difference $y^c - y^o$ between calculated and observed values states directly whether calculated absorption is more or less exothermic than measured. A differential absorption heat is $(\partial H_{\mathrm{eq}}/\partial n_{\mathrm{CO_2}})_{T,P,n_A,n_w} - h^{\mathrm{feed}}_{\mathrm{CO_2}}$; it is distinct from the finite-loading integral. Each integral comparison retains its initial stock, final loading and any imposed liquid-loading cap. For an unloaded binary mixture, excess enthalpy per mole of mixture is

$$H^E = h_{\mathrm{mix}} - x_w h^*_w - x_A h^*_A, \qquad \rho_{\mathrm{mass}} = \frac{m_{\mathrm{feed}}}{V_{\mathrm{eq}}}. \tag{26}$$

The superscript $*$ denotes the pure-component liquid at the same temperature and pressure as the mixture; the ideal-gas enthalpies cancel in H^E. Density uses the complete analytical mass, including added CO_2 for loaded solutions, and the equilibrium liquid volume. Heat capacities use the same complete-feed molar or mass basis as their source observations.

3.2. Parameters and fitting

Each bounded least-squares fit minimizes its original weighted sum,

$$C = \frac{1}{2}\boldsymbol{r}^{\mathsf{T}} W \boldsymbol{r}, \qquad r_j = (y^c_j - y^o_j)/s_j \quad \text{or} \quad r_j = \ln(y^c_j/y^o_j)/s_j. \tag{27}$$

Here y^c_j and y^o_j are calculated and observed values, s_j denotes the assigned observation scale and W includes group weights or correlations. The objectives use different observations and scales, so their costs do not measure relative physical accuracy across amines.

For MDEA, Kuranov total pressures use natural-log ratios, scale one and weight $100/9$; Kim binary total pressures use natural-log ratios with unit scale and weight. Jakobsen protonated fractions use differences scaled by 10 % of the observed fraction, with unit weight. Carson and Merkley heat differences use 3000 J/mol CO_2 scales and unit weights. Weiland heat capacities use 2 % of the observed heat capacity and unit weights. Loaded-density differences use the observation-specific scales and weight $1/33$, whereas unloaded densities use 0.5 kg/m^3 and weight ten.

AMP loaded total pressures use log ratios with unit scale and weight $100/[9(47)\max(u_P/P, 0.05)^2]$. The fifty-two binary entries comprise twenty-six pressures and twenty-six vapor AMP fractions; binary log pressures use 0.05 scale through their weight, while vapor fractions use $\max(10^{-4}, 0.05y)$. The binary group is averaged over its fifty-two entries. Ciftja carbamate fractions are mapped to carbamate per total amine, with difference scale 0.003 and group weight $1/57$.

PZ pressure differences use an assigned scale 0.02 times observed pressure. NMR normalization propagates independent area errors with $q = 0.02$ into correlated group-fraction blocks; complete-state areas reconstruct the monocarbamate pair and free/dicarbamate groups before normalization. Blank dicarbamate observations are not zeros and do not create additional residuals. The assumed area scale is not a source-validated independent species standard deviation.

For DEA, pressure log ratios use scale 0.09 and heat differences use 0.02 times the magnitude of measured heat, with each group's squared residuals averaged.

The average absolute relative deviation and signed bias are

$$\mathrm{AARD} = \frac{100\,\%}{n}\sum_{j=1}^{n}\left|y_j^c/y_j^o - 1\right|, \qquad \mathrm{bias} = \frac{100\,\%}{n}\sum_{j=1}^{n}(y_j^c/y_j^o - 1). \tag{28}$$

The same reporting formula is applied separately to nonzero heats and other properties; species differences and root-mean-square errors retain their observable units. A negative pressure bias means underprediction.

The AMP results use the fit whose carbamate diameter is bounded below by the MEA carbamate diameter, 5.32 Å; a fit allowed down to 3.5 Å lowers the cost only slightly, so the data identify the diameter only weakly below about 5.3 Å, and that fit is a sensitivity case. The PZ fit stops at its 100-iteration limit with nearly flat cost and is not labeled formally converged. DEA's two charged-chain starts differ in cost below the optimizer tolerance and represent a numerical tie rather than distinct physical optima. The selected MDEA fit was chosen partly for its held-out pressures and gives poorer agreement with Carson's heats than the earlier fit (scaled root-mean-square (RMS) residual 1.106 against 0.977). The selected parameters remain fixed for the reported comparisons; Supplementary Section S3 lists the starting points, bounds and alternative fits.

Local covariance holds shared inputs, model choices, residual weights and bound fitted parameters fixed. For interior fitted parameters, the covariance is the local inverse weighted information matrix multiplied by the retained residual-variance factor. At a prediction with parameter derivative g, the conditional variance is $g^{\mathsf{T}}C_\theta g$; displayed intervals use ± 1.96 conditional standard errors (SE), not total prediction intervals. Separate sensitivities displace one marginal SE at a time and do not include correlation compensation. Inward profiles fix a bound fitted parameter 1, 5 or 10 % of its original interval inside the bound and refit the remaining fitted parameters. Those displacements have different meanings and sizes; none quantifies source systematics or model-form error.

3.3. Phase-stability search

At each parent liquid, a trial phase minimizes the tangent-plane distance $\sum_i n_i^{\mathrm{trial}}(\mu_i^{\mathrm{trial}} - \mu_i^{\mathrm{parent}})/(RT)$ under reaction equilibrium and electroneutrality. A negative distance is followed by a material-conserving split calculation; only a Gibbs-energy decrease identifies an additional phase. Unlike the penalized MEA fit of Part I [1, Sect. 4.5], the reported MDEA, AMP, PZ and DEA fits contain no stability penalty; their phases were searched after fitting. A separate PZ fit applies that penalty and is reported only to show its remaining false liquid splits and the finite search's missed solutions. All split counts are lower bounds; absence of a detected split is not proof of stability. The sampled states and unavailable states remain distinct from a continuous phase boundary.

4. Results and discussion

4.1. Subsystems

The binary subsystems agree closely for MDEA but retain discrepancies for PZ and DEA. MDEA binary-pressure AARD is 3.23 % (Table 12). The separately fitted PZ–water binary pressures and vapor compositions give scaled root-mean-square residuals of 7.78290 and 12.75239, respectively, using a pressure scale of 0.3 kPa and the conditional vapor-composition scale $\sqrt{(0.01y)^2 + (0.00005/\sqrt{3})^2}$; these discrepancies are unchanged by the reactive refit. For water–DEA, the comparison-only Cai binary data [69] give 40.82 % pressure AARD at 6.66 kPa, far from the fitted binary isotherm, and the calculated water–DEA excess enthalpy has the wrong sign (Section 4.2.3).

4.2. Loaded aqueous amines

Amine-specific fitting recovers parts of the targeted behavior, but the remaining deviations are larger than several of the assigned observation scales (Table 12). MDEA fitted Kuranov total-pressure AARD is 20.44 %. Its source-scaled RMS exceeds unity for protonation fraction, both heat sources, heat capacity and both density groups. For Carson heat, scaled RMS increases from 0.977 with the earlier shared description to 1.106 after refitting to the Part I shared description. This compromise accompanies recovery of held-out pressures rather than uniformly improved properties.

Table 12
Calibration deviations. AARD uses each observable separately. Scaled RMS is the root mean square of residuals divided by their assigned observation scales, before applying additional objective weights. Pressure residuals are logarithmic for MDEA and relative differences for PZ. These statistics do not provide a cross-amine accuracy ranking.

Observable and fitted source	n	AARD / %	Scaled RMS
MDEA: Kuranov total pressure	67	20.44	0.22
MDEA: protonated fraction	16	13.58	1.64
MDEA: Carson heat	40	5.45	1.11
MDEA: Merkley heat	4	8.46	2.02
MDEA: C_P	33	2.59	1.76
MDEA: loaded density	33	0.18	1.59
MDEA: unloaded density	24	0.06	1.44
MDEA: binary pressure	61	3.23	0.04
AMP: Hartono total pressure	47	10.33	–
PZ: Hilliard partial pressure	62	19.16	12.90
DEA: Lawson partial pressure	30	28.63	–
DEA: Carson integral heat	50	0.98	–

4.2.1. CO_2 solubility

Small aggregate biases in the fitted AMP, PZ and DEA pressures conceal loading-shape errors (Fig. 1). AMP fitted total-pressure AARD is 10.33 %, with bias −0.90 %. The fitted PZ Hilliard partial pressures give 19.16 % AARD and −3.23 % bias, but RMS is 12.9 times the assigned pressure scale.

DEA fitted pressure AARD is 28.63 %, with bias −0.76 %, whereas integral-heat AARD is 0.98 %. Only 10 of 30 Lawson pressures satisfy $|\ln(p^c/p^o)| \leq 2(0.09)$, twice the assigned log-pressure residual scale, and each isotherm's highest-loading pressure is 31–58 % low.

Unfitted pressure errors change with source and temperature; the largest occur in extrapolated blocks, but for MDEA, AMP and DEA they already reach 28–52 % AARD at conditions inside the measured law ranges. Held-out MDEA partial-pressure AARD is 28.19 % over twenty Xiao observations at 23.5 wt% and 298.15–313.15 K, all inside the dissociation law's measured range. Every Xiao pressure is underpredicted. The Ermatchkov aggregate, 18.27 % over eighty-eight observations at no more than 49 wt%, mixes measured-law and extrapolated conditions. Inside 278–368 K, its sixty-two pressures give 16.31 % AARD and −11.59 % bias; the twenty-six pressures at 393.15 K give 22.94 % and +21.93 %, respectively. Fitted Kuranov total pressures exhibit the same sign reversal: 11.80 % AARD with −7.80 % bias inside the range, and 31.09 % with +31.09 % bias above it (Table 13). This is an observation about the calculated results, not a diagnosis of the dissociation law; no calculation isolates its contribution from the other temperature-dependent terms.

Table 13
Fitted MDEA errors separated against the 278–368 K dissociation measurements. Heat signed errors are calculated minus measured; they are not relative biases.

Observable	Law range / T	n	AARD / %	Bias / signed error
Kuranov total pressure	inside	37	11.80	−7.80
Kuranov total pressure	outside	30	31.09	31.09
Merkley heat (20 and 40 wt%)	inside (333.15 K)	2	10.36	-5725 J/mol
Merkley heat (20 and 40 wt%)	outside (388.71 K)	2	6.56	-3989 J/mol

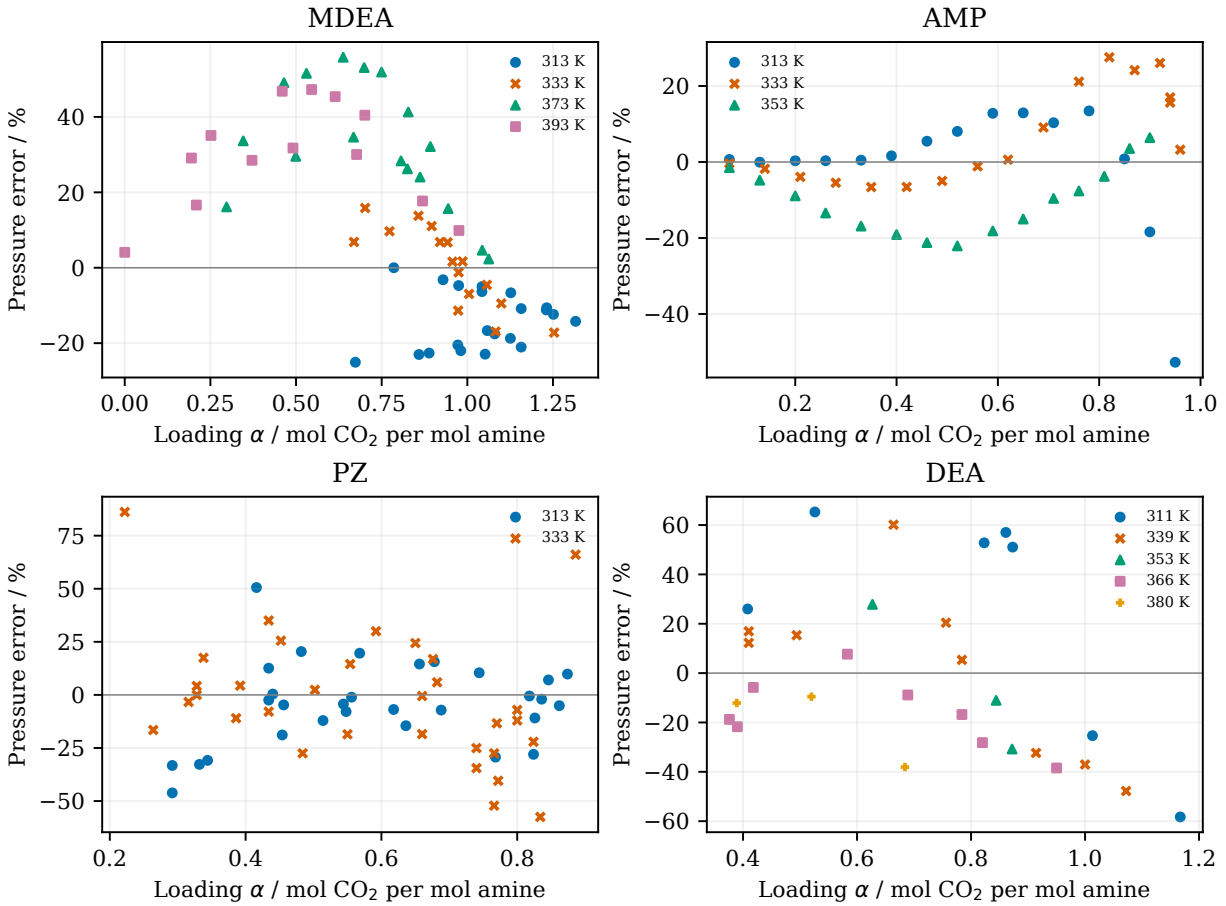


Figure 1: Fitted pressure errors against analytical loading. MDEA and AMP use total pressure; PZ and DEA use CO_2 partial pressure. Each point is a calculation at a measured state. Colors distinguish source temperatures within each panel; no model curve is inferred from the discrete points. MDEA points at 373 and 393 K lie above the 368 K limit of the dissociation measurements.

Table 14
Unfitted pressure errors by source and temperature. MDEA sets were previously accessed; AMP Kundu and PZ Ermatchkov were scored before final parameter selection, and DEA was first scored on a numerically tied fit. Kamps was scored once after the PZ fit was frozen. All are partial pressures except AMP Tong and PZ Kamps.

Amine and source	Conditions	*n*	AARD / %	Bias / %	Measured-law range
MDEA Xiao	all	20	28.19	−28.19	inside
MDEA Ermatchkov	all	88	18.27	−1.68	mixed
MDEA Ermatchkov	278–368 K	62	16.31	−11.59	inside
MDEA Ermatchkov	above 368 K	26	22.94	21.93	dissociation extrapolated
AMP Kundu	all	51	42.85	37.79	mixed
AMP Kundu	303.0 K	25	53.94	53.94	hydrolysis slope extrapolated
AMP Kundu	313.0 K	15	43.78	43.78	inside
AMP Kundu	323.0 K	11	16.35	−7.09	inside
AMP Tong (total)	313–353 K	17	28.66	8.11	inside
PZ Ermatchkov	all	52	21.94	7.82	mixed
PZ Ermatchkov	313.1 K	19	9.32	−8.70	inside
PZ Ermatchkov	353.15 K	18	13.12	−8.64	carbamate extrapolated

Amine and source	Conditions	*n*	AARD / %	Bias / %	Measured-law range
PZ Ermatchkov	393.15 K	15	48.52	48.52	carbamate and protonation extrapolated
DEA Seo & Hong	all	16	79.54	77.80	mixed
DEA Seo & Hong	40 °C	5	28.51	22.95	inside
DEA Seo & Hong	60 °C	6	59.82	59.82	protonation extrapolated
DEA Seo & Hong	80 °C	5	154.22	154.22	protonation extrapolated
DEA Lal	all	44	44.17	−22.93	mixed
DEA Lal	40 °C	17	52.25	−48.61	inside
DEA Lal	100 °C	27	39.09	−6.76	protonation extrapolated
PZ Kamps (total)	all	94	42.5	−40.3	mixed; comparison-only

The reported AMP fit gives Kundu AARD 42.85 % and bias +37.79 % over all fifty-one observations. The signed error decreases with temperature: +53.94 % at 303 K, +43.78 % at 313 K and −7.09 % at 323 K. The 303 K set extrapolates the source basis of the hydrolysis slope below 313 K; all three temperatures are within the fixed dissociation law's interval. Tong's seventeen total pressures [70] at 313–353 K give 28.66 % AARD and +8.11 % bias, while twenty-three rows at 373–393 K lie above the dissociation law's 353.22 K calculation limit and are not calculated. The Tong biases also decrease with temperature, from +48.55 through +5.73 to −23.19 % at approximately 313, 333 and 353 K. The decreasing signed errors and bound hydrolysis enthalpy correction are relevant to temperature dependence, but do not distinguish model error from disagreement between experiments.

PZ Ermatchkov partial pressures [12] give 21.94 % AARD overall, with strong temperature dependence. At 313.1 K, within the carbamate measurements, AARD is 9.32 %; at 353.15 K, above those measurements, it is 13.12 %; at 393.15 K, above both carbamate and protonation measurements, it reaches 48.52 %. Only eleven of fifty-two pressures lie within the printed source uncertainty, and the uncertainty-scaled RMS is 11.59. The mean error consequently understates the disagreement relative to the source's measurement resolution.

The comparison-only total pressures of Pérez-Salado Kamps et al. [11, Table 3, p. 2666] were scored once on the frozen PZ fit with twelve fitted parameters and never fitted or used to select it. The ninety-four rows give 42.5 % AARD and −40.3 % bias. Loaded rows below 0.2 MPa give 20.4 % AARD and −6.0 % bias; the loaded 374–395 K rows at or above 0.2 MPa give 37.1 % and −35.2 %, and the remaining loaded 313–354 K rows give 53.0 % and −53.0 %. The 313.75 K block, inside both the 283.15–333.15 K carbamate and 293.14–353.42 K protonation measurements, still has −32.1 % bias, so leaving those ranges is not the only cause. Only that block lies inside both PZ laws; the 395.0 K rows also exceed the shared R2 interval (Supplementary Table S11).

The source gives pressure bounds of ±4 or ±10 kPa, without assigning the transducer to each row, and uncertainty up to 10 % below 0.1 MPa [11, pp. 2664, 2667]. All eighty-two loaded rows at or above 0.2 MPa lie outside both pressure bounds. The ±0.7 % solvent-mass bound moves calculated pressure by up to −10.8 % / +11.7 %; the median ratio of sensitivity to misfit is 0.07 and the maximum is 0.88. Near 1 mol/mol at low pressure this sensitivity can be comparable to the misfit, but for thirty rows above 1.2 mol/mol and at or above 0.2 MPa it is at most 3.2 % against misfits of at least 34.9 %. These are input sensitivities, not propagated total uncertainties; the individual charged CO_2 masses needed to propagate their uncertainty were not reported.

DEA predicts Seo and Hong's thirty-mass-percent solution [71] with 79.54 % AARD and +77.80 % bias; the 80 °C subset reaches 154.22 %. Lal's more dilute stock [72] instead gives 44.17 % AARD and −22.93 % bias. Even the 40 °C subsets, within the Bower measured temperature range, have AARDs 28.51 % against Seo and Hong and 52.25 % against Lal. The higher-temperature subsets extrapolate that law, with the source-specific errors listed in Table 14. Lal's low-pressure observations, which the source reports as less accurate, remain in the AARD (Supplementary Section S4). Lawson's comparison-only Table VIII pressures, nominally 50 wt% but 54.5 wt% at 77 °F, give 41.74 % AARD.

The source tables indicate a possible contribution from laboratory disagreement in DEA. At loading about 0.41, Lawson reports 16.0–16.7 kPa at 65.6 °C and 25 wt%, while Seo and Hong report 11.6 kPa at 80 °C and 30 wt%. These are not matched states: temperature, concentration and loading differ. They do not isolate experimental bias or establish which source is correct, and the model retains a loading-shape error within its own fitted source. Across amines, unmatched pressure types, source uncertainties, concentrations, temperatures and selection histories prevent comparison of their error percentages as an accuracy ranking.

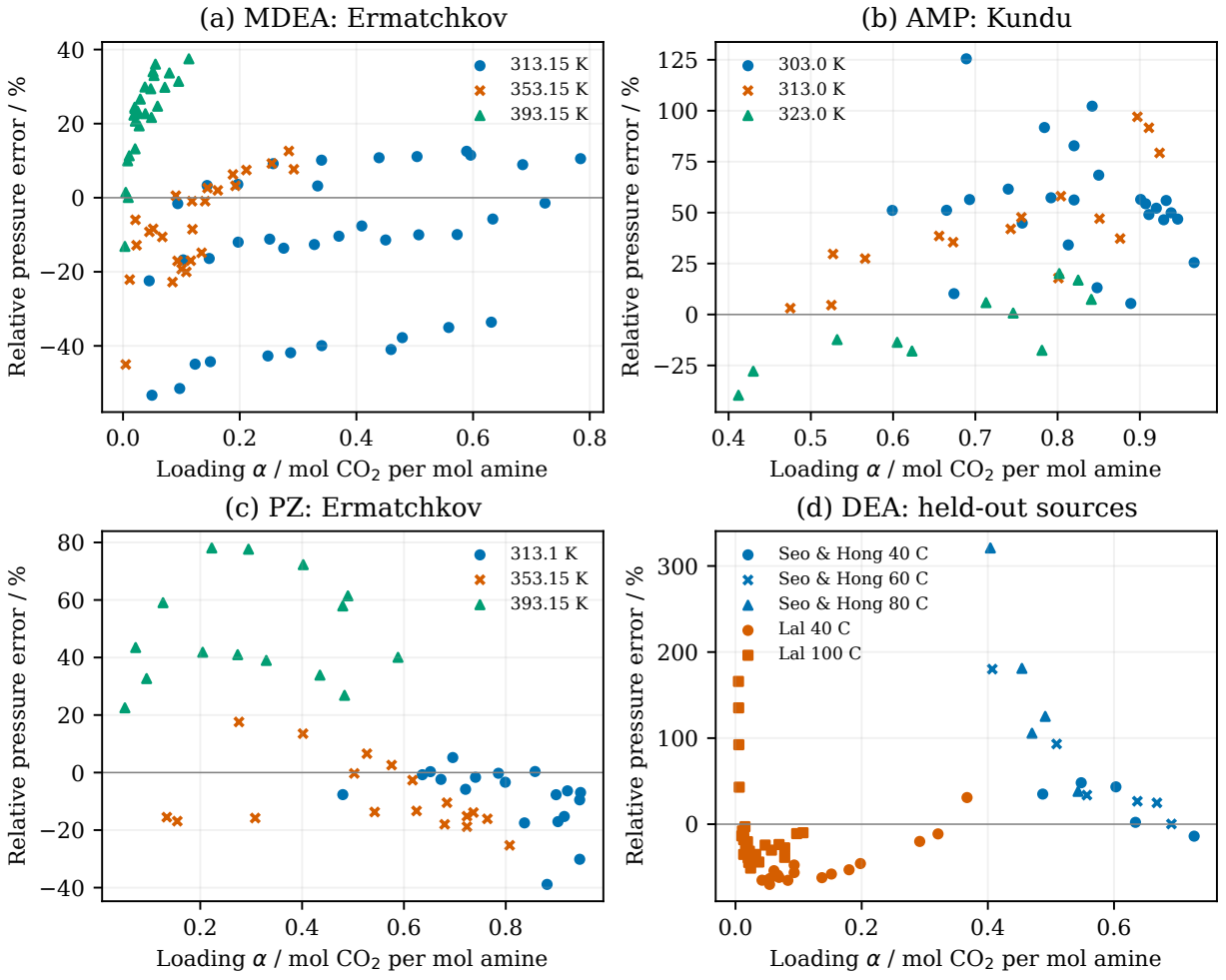


Figure 2: Relative partial-pressure errors at measured loadings. All panels use held-out sets; the MDEA set was previously accessed. MDEA at 393.15 K exceeds its dissociation measurements; PZ at 353.15 K exceeds its carbamate measurements and at 393.15 K also its protonation measurements; AMP at 303 K extrapolates the hydrolysis source slope. DEA marker shapes distinguish 40, 60, 80 and 100 °C. Points from different stocks are shown without interpolating a continuous relation.

Temperature-range separation narrows the interpretation without assigning cause. MDEA and PZ have larger positive errors in high-temperature blocks that extrapolate measured reaction laws, but MDEA Xiao and DEA's low-temperature held-out subsets already show substantial errors inside those law ranges. AMP's decreasing signed error with temperature occurs while its fitted hydrolysis enthalpy correction is on a limit. These observations indicate conditions where the thermodynamic description is inaccurate; they do not isolate a law slope from an interaction, a solvent-property extrapolation or source disagreement.

The source comparisons also remain incomplete. AMP lacks sufficient matched experimental states to separate the Kundu/Hartono difference from concentration-dependent model error, especially at its coldest isotherm. Gas dilution, assumed cell pressures, PZ heavy-water treatment and omitted byproduct chemistry have distinct origins and should not be collapsed into one explanation for all pressure errors.

4.2.2. Liquid speciation

PZ species residuals remain large relative to their assumed measurement scale, although the fitted PZ pressure AARD is moderate. PZ mean absolute monocarbamate- and dicarbamate-group errors are 0.0258 and 0.0115 per total PZ, and the NMR whitened RMS remains 11.58 under the assumed area-error model. A moderate mean relative

pressure error does not mean agreement at source resolution or an adequate species fit. For AMP, the fitted carbamate-fraction RMS is 0.00631.

Comparison-only DEA species fractions agree closely with Böttinger's measurements. At nine Böttinger states without the oxazolidone byproduct, mean absolute true-mole-fraction differences are 0.001749 for DEA plus $DEAH^+$, 0.001301 for carbamate and 0.001274 for the bicarbonate-labeled sum. The latter is compared with model bicarbonate plus carbonate. Sixteen byproduct-containing states remain separate because that chemistry is omitted, and no NMR mole-fraction measurement uncertainty is reported. The pressure assumption and aggregate species definitions condition the comparison (Fig. 3(b)).

4.2.3. Absorption heat and heat capacity

Close agreement in fitted DEA absorption heats does not extend to the water–DEA mixing enthalpy, which has the wrong sign. Forty-six of the fifty Carson heats lie within the assigned 2 % scale (Fig. 3(a)). Heat agreement is restricted to these 298.15 K measurements and does not resolve the independent mixing-heat discrepancy below.

At 363.15 K and 1.75 MPa, all fourteen water–DEA excess-enthalpy calculations [54] have the wrong sign: approximately +86 to +758 J/mol, against measured values from −93 to −1707 J/mol. None lies within twice the source scale. This error concerns heat of mixing per mole of water–DEA mixture, whereas Carson's fitted heats concern absorption per mole of added CO_2 at 298.15 K. Agreement with the latter cannot establish a correct mixing enthalpy at another temperature. No corresponding excess-enthalpy comparison is available for MEA, MDEA, AMP or PZ, so the sign error has not been shown to be DEA-specific. Cancellation of mixing contributions during absorption is possible but was not calculated.

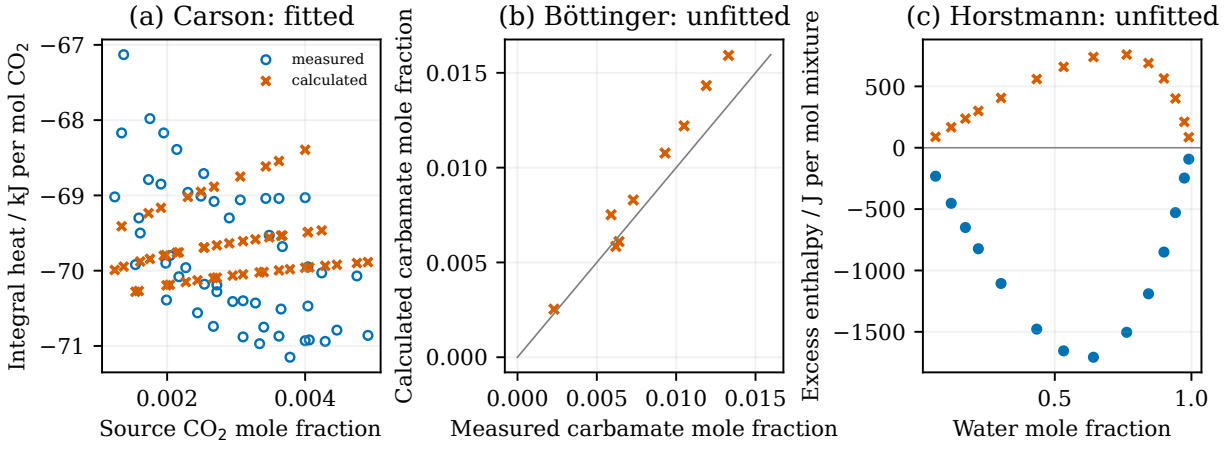


Figure 3: DEA properties at the measured states: (a) fitted Carson integral absorption heats at 298.15 K against the source CO_2 mole fraction, with the source heat-release magnitude converted to negative enthalpy for exothermic absorption; (b) comparison-only carbamate mole fractions in the nine Böttinger states without byproduct, at assumed 2.5 MPa; (c) comparison-only water–DEA excess enthalpy at 363.15 K and 1.75 MPa. Circles are measurements and crosses calculations. The different enthalpy bases and temperatures separate the successful absorption-heat calibration from the wrong-sign mixing prediction.

The MDEA held-out capped absorption heat at 388.71 K and 60 wt% is −64.491 kJ/mol CO_2, against −63.3 kJ/mol; this extrapolates the dissociation measurements and solvent concentration. The corresponding 333.15 K, 60 wt% capped heat could not be calculated. PZ has no comparable reactive heat assessment; the four amines have not been checked equally across caloric properties.

4.2.4. Density

Density was compared for MDEA and DEA only. Forty-three comparison-only loaded DEA densities at 298.15 K give 0.445 % AARD (Supplementary Section S4); the fitted MDEA density groups are given in Table 12. PZ has no comparable reactive density assessment; the four amines have not been checked equally across volumetric properties.

4.3. Fitted parameters, conditional uncertainty and sensitivity to the shared description

4.3.1. Conditional parameter uncertainty

For AMP and DEA, conditional parameter bands are far smaller than the observed unfitted-pressure errors (Fig. 4). AMP Kundu relative SEs range from 3.2 to 13.5 %, with median 6.7 %; only eight of fifty-one observations fall within the nominal ±1.96 SE intervals. DEA relative pressure SEs are 5.4–11.6 % for Seo and Hong and 12.3–19.1 % for Lal, compared with their much larger measured deviations. These comparisons show that uncertainty within the selected local parameter fit does not encompass the discrepancies. They do not identify which experiment or model term causes a miss.

MDEA bands at the 108 Xiao and Ermatchkov pressure states have relative SEs of 7.82–39.1 %, overlapping the magnitude of the held-out errors. Its cation–bicarbonate interaction and cation Born diameter lie at +1 and 7 Å, respectively, and remain fixed in this covariance. For AMP, the carbamate diameter and hydrolysis enthalpy correction are fixed at their limits, 5.32 Å and −5000 J/mol. The corresponding residual-variance factors are 2.7639784 for MDEA and 0.829 for AMP. The band size therefore depends on both the fitted information and the conditional error model, not just the parameter count.

PZ's local linear uncertainty is less reliable: one marginal SE of a fitted dispersion energy changes some Hilliard calibration pressures by 90–160 % in the linear estimate. Its cation–bicarbonate interaction is -0.02 ± 1.03 across an allowed interval of width 1.5, indicating weak identification. With the PZH^+–water interaction fixed at its −1 limit, eleven interior fitted parameters use variance factor 146.3; calibration-pressure relative SEs range from 2.8 to 34 %. No PZ held-out pressure band was calculated, and the calibration band cannot substitute for it. The inward 5 and 10 % profile refits did not converge, so their costs are upper bounds.

DEA has no active parameter bounds, but its hydrolysis-law level and carbamate–water interaction have correlation 0.972. Its residual-variance factor is 0.210, and relative SEs at fitted Lawson and Carson states are 3.3–10.3 % and 0.9–1.9 %, respectively. Small conditional errors coexist with compensation between parameters. None includes shared-input uncertainty, experimental systematic errors, uncertainty in the omitted chemistry or model discrepancy.

The amines differ in parameter freedom and observations. PZ needs a more extensive species network, retains a weak cation–bicarbonate interaction and large local energy sensitivities, and, as Section 4.5 shows, predicts false additional liquids. DEA's four reactive parameters have small conditional errors and no active bounds, yet large unfitted-pressure errors remain. AMP's carbamate diameter and enthalpy correction reach limits with nearly flat inward profiles (Supplementary Section S6). A diameter at a bound is evidence about this constrained fit, not a measurement of an ion size. One solute per chemical class, different data objectives and different phase grids do not distinguish an intrinsic class effect from those other changes.

4.3.2. Sensitivity to the shared description

Unfitted pressures depend on the shared description as well as on the amine fit: substituting Part I's bicarbonate–water interaction and R2 law moves the MDEA and AMP errors by 22–26 percentage points in opposite directions before the amine parameters are refitted (Table 15). Table 15 follows the same data through the earlier shared description, substitution of the Part I bicarbonate–water interaction and fitted R2 law, and amine-specific refitting. For AMP, Kundu AARD changes from 43.69 to 21.89 % upon substitution without refitting, then returns to 43.51 % when the original AMP fitted parameters and limits are refitted. The substitution worsens fitted total-pressure AARD to 18.85 %, while refitting recovers 9.93 %. The favorable 21.89 % score thus reflects cancelling changes in this comparison, not a jointly improved calibration and independent pressure description.

Table 15
Same-data pressure AARD (%). Each row compares the earlier shared inputs, substitution of the Part I bicarbonate–water interaction and R2 law without refitting the amine, then amine-specific refitting. A3 is a counter-case, not the selected MDEA description. AMP here uses the original bounds; its final bounded fit is reported separately.

Amine fit (unfitted source)	Earlier	Substituted	Refitted
MDEA alternative ion description A3 (Ermatchkov)	26.21	52.47	31.44
MDEA reported fit (Ermatchkov)	19.15	42.03	18.27
AMP (Kundu)	43.69	21.89	43.51

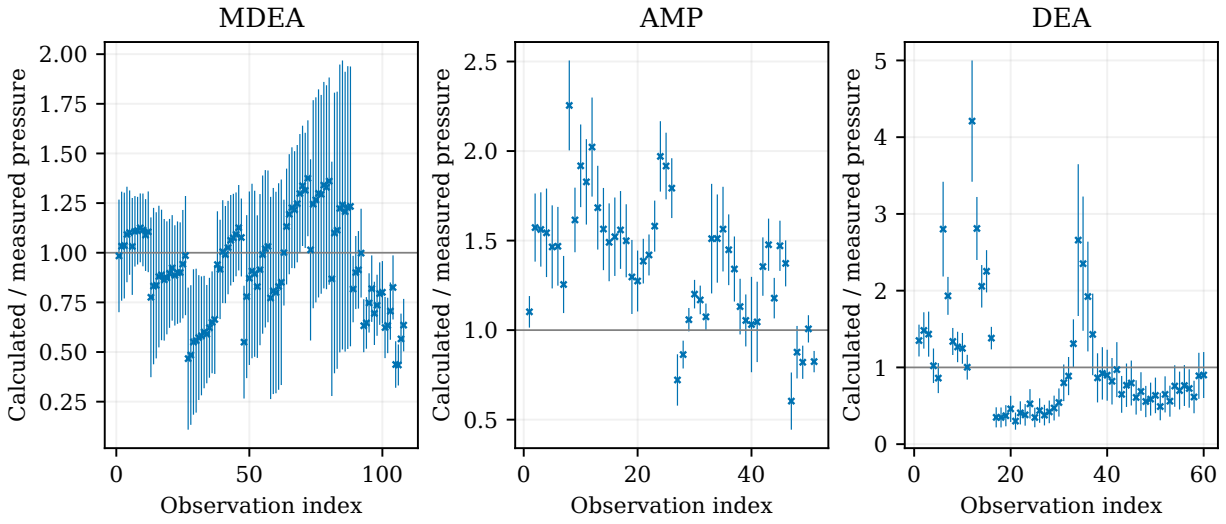


Figure 4: Calculated-to-measured partial-pressure ratios with nominal ±1.96 conditional parameter-SE intervals, at the measured states. Separate panels show MDEA Xiao/Ermatchkov, AMP Kundu and DEA Seo and Hong/Lal; the horizontal axis numbers the observations within each panel and is not a temperature or loading axis. The horizontal line is equality. Intervals are linear parameter propagation, conditional on fixed shared inputs and active bounds; they exclude measurement noise and model discrepancy. PZ held-out bands are unavailable.

For the reported MDEA fit, the same sequence changes Ermatchkov AARD from 19.15 to 42.03 and then 18.27 %. An alternative MDEA ion description, A3, changes from 26.21 to 52.47 and then 31.44 %, so refitting does not recover its earlier error. A3 is a counter-case, not an additional selected model. These same-source interventions demonstrate parameter compensation. For MDEA, which tests the shared inorganic-carbon description without an amine-carbamate pathway, the fitted cation–bicarbonate interaction also ends on its +1 bound; this bicarbonate-related pattern does not uniquely assign the discrepancy to bicarbonate, Born solvation or a reaction constant.

The reported AMP fit widens the cation–CO_2 interaction and retains the 5.32 Å carbamate floor, giving 42.85 % Kundu AARD. Lowering the floor to 3.5 Å gives a slightly smaller calibration cost but 43.95 % Kundu AARD; it is not the reported result. A model-free comparison of the Hartono and Kundu measurements lacks sufficient matched conditions, especially at 303 K, to decide whether Hartono and Kundu disagree or whether the model misses their concentration dependence. Its interpolation and vapor-baseline assumptions do not establish that either source is erroneous.

4.4. Comparison with published models

Table 16
Published-model comparisons with data roles stated. Only the PZ Hilliard row has the same fitted source and row count in both models; objective and chemistry still differ. Tong coverage differs. MDEA and qualitative comparisons use different observation subsets.

Model / system	Source and published deviation	Published role / role here
Yi e-NRTL, PZ [9, Table 3]	Hilliard: 62 points, 18.0% partial-pressure AARD	Fitted / fitted
Yi e-NRTL, PZ [9, Table 3]	Ermatchkov: 52 points, 7.8% partial-pressure AARD	Fitted / held out; previously scored
Yi e-NRTL, AMP [9, Table 4]	Kundu: 51 points, 15.4% partial-pressure AARD	Fitted / held out; previously scored
Yi e-NRTL, AMP [9, Table 4]	Tong: 45 points, 8.0% total-pressure AARD	Fitted / unfitted; 17 points here

Model / system	Source and published deviation	Published role / role here
Wangler ePC-SAFT, MDEA [22, Table 11]	Own 82-point set: ARD 19.7%; corrected Uyan 34.4%	Ternary predictions / different reactive fit and partial-pressure subsets here
Cleeton ePC-SAFT, MDEA [23, Sect. 3.3.2]	Concentration and low-loading comparisons; no matched score	Ternary predictions after pure/binary estimation
Perdomo SAFT-γ Mie, PZ / DEA [21, Sects. 4.2, 4.3.3]	PZ concentration transfer; Lal pressures and Böttinger species	Selected reactive group estimation; other concentrations/species predicted
Nasrifar PC-SAFT, DEA [17, Sect. 6, Figs. 13–14]	Species and acid-gas curves; no matched score	Predicted after pure/binary estimation; ideal-activity chemistry

Only the Hilliard PZ pressures are fitted both here and in a published model, where the eNRTL deviation of Yi et al. is similar (18.0 against 19.16 %); every other published comparison differs in which observations were fitted.

Yi et al.'s eNRTL model provides two same-source PZ comparisons: 18.0 % partial-pressure AARD for the same sixty-two Hilliard observations, fitted in both models, and 7.8 % for fifty-two Ermatchkov observations, fitted by Yi but held out here [9, Table 3, p. 2]. For AMP, Yi reports 15.4 % on the fifty-one Kundu partial pressures, fitted there and held out here [9, Table 4, p. 5]. Yi neglects AMP carbamate [9, Sect. 2.1.2, p. 3], whereas the present fit includes carbamate and fits its species fraction. Yi's 8.0 % Tong result is total-pressure AARD on forty-five fitted observations, compared with seventeen unfitted total pressures scored here. That unequal coverage is not a matched-data performance comparison.

Wangler et al. report MDEA pressure ARD 19.7 % over their eighty-two ternary observations and correct Uyan's result to 34.4 % on that set [22, Table 11]. Both are ternary predictions after lower-order estimation, including osmotic data for the ion–water interactions in Wangler's work; neither fits those ternary pressures. The present MDEA fit includes reactive total pressures, and its held-out partial-pressure subsets are different, so those percentages cannot establish relative accuracy. Cleeton et al. give no matched score; their separation of data roles and condition-dependent errors [23, Sect. 3.3.2, Figs. 7–8] is more informative here than a pooled score from unmatched sets.

Of the models reviewed in Section 1.1, SAFT-γ Mie predicts DEA pressures from Lal and species from Böttinger after group-interaction estimation [21, Sect. 4.3.3, Figs. 13–14], and Nasrifar and Tafazzol predict DEA acid-gas curves after pure and binary estimation with ideal-activity chemistry [17, Sects. 2–6, Figs. 13–14]. These qualitative comparisons distinguish fitting policies and chemistry; they supply no matched-row ranking against the present DEA fit.

4.5. Phase stability

Small property residuals do not exclude false liquid phases: the reported PZ fit has at least forty across calibration and comparison states, and a separate fit with the Part I stability penalty still has at least three, two of which the standard finite search missed.

Four false liquid splits are detected for the reported PZ fit at Hilliard rows 55, 56, 61 and 62, near 4.96–5.02 mol PZ/kg water, loading 0.77–0.83 and 313 or 333 K. The calculated additional liquids contradict the measured liquid states despite the moderate pressure AARD. The search examines 183 calibration states, comprising 121 NMR states and sixty-two bubble states, and finds no split at the fifty-two held-out bubble states. Rich NMR states below their calculated bubble pressure are assessed as liquids at 1.05 times that pressure; those moved states do not demonstrate single-liquid behavior at the original assumed 0.1 MPa.

Thirty-six additional false liquid splits occur at Kamps states, at 313–354 K and loading 0.96–1.64 mol CO_2/mol PZ, with tangent-plane distances from -0.026 to $-0.554\ RT$. The second liquids are rich in bicarbonate and doubly protonated PZ. Pressures at these states belong to a single liquid that is not the stable state of this model; the split count is a lower bound. At 1.5 times bubble pressure all four Hilliard splits persist, although the standard finite search misses two; all 121 NMR states pass the finite search at $T + 0.5$ K, with rich states moved above bubble pressure as specified in the supplement.

The Kamps states combine these false splits with the underprediction inside both PZ law ranges reported in Section 4.2.1. At the thirty-six Kamps split states, doubly protonated PZ accounts for 2–29 % of total PZ in the calculated parent liquid, whereas the source's model describes it as present "only in very small amounts" at 353 K and about 4 mol/kg [11, p. 2666]. This is a model-speciation observation, not a measured fraction. Candidate causes,

without ranking, are physical CO_2 solubility at MPa pressure, PZH^+–bicarbonate and zwitterion interactions, second protonation through its law and tied PZH_2^{2+} parameters, and the shared MEA CO_2–water description. This comparison cannot separate them.

The separate PZ penalty fit does not remove the false liquid splitting. Its standard finite search detects only one split, but direct tangent-plane evaluations at known split compositions show two further negative distances, $-0.0309\,RT$ and $-0.0437\,RT$, at rows 55 and 56. The penalty fit therefore has at least three false liquid splits. This counterexample demonstrates a missed finite search, so every split count in this paper is a lower bound, and the absence of detected instability, here and for the penalized MEA fit of Part I [1, Sect. 4.5], does not prove stability.

The MDEA searches cover 775 calculated states, some repeating a physical state; none detects an additional liquid, 37 detect a gas-like phase, and none is ambiguous or unresolved. Seven source states above the shared R2 interval are excluded, and there is no phase evidence at 394.94–395.00 K. AMP has no detected liquid, gas, ambiguous or unresolved outcome at the fifty-one Kundu states. Its calibration searches retain no detected liquid, ambiguous or unresolved outcome, but their gas outcomes were not recorded. DEA searches the 140 fitted and held-out states without detecting an additional phase or an unresolved result.

Table 17
Finite phase-search results. Detected liquid-split counts are lower bounds.

Fit / states	Detected liquid splits	Other outcomes and omissions
MDEA: 775 searched states	0 found	37 gas-like phases; no ambiguous or unresolved result; seven states above the R2 interval not calculated
AMP: 51 Kundu states	0 found	No gas, ambiguous or unresolved result; calibration gas outcomes not recorded
PZ: 183 calibration states	at least 4	No ambiguous or unresolved result; three rich NMR pressures moved above bubble pressure
PZ: 52 held-out bubble states	0 found	Finite search only
PZ: 94 Kamps bubble states	at least 36	56 without detected split; 2 unloaded undecided; no gas or ambiguous split
PZ: compressed Hilliard states	at least 4	At 1.5 times bubble pressure; routine search misses two
PZ: 121 NMR states at $T + 0.5$ K	0 found	Three rich states above 0.1 MPa evaluated at 1.05 times bubble pressure
PZ penalty alternative	at least 3	One found routinely; two additional splits resolved by direct distance evaluation
DEA: 140 fitted/held-out states	0 found	No gas, ambiguous or unresolved result

4.6. Scope of the evidence

The common equation of state, reference conventions and frozen water/carbonate description can be used with different amine reaction networks while fitting only amine-specific quantities. What transfers is that thermodynamic description and a conditional ability to fit selected properties; uniform pressure, caloric and phase accuracy does not follow.

Within the tested conditions, AARD is 0.98 % for fitted DEA integral absorption heats at 298.15 K, 0.445 % for comparison-only loaded DEA densities at 298.15 K, and 9.32 % for held-out PZ partial pressures at 313.1 K. Tests at the PZ false splits (Hilliard loading 0.77–0.83 and Kamps loading 0.96–1.64), for the water–DEA mixing enthalpy and above reaction-law measurement ranges reveal the limitations reported here; conditions beyond the measured and searched states were not tested. The comparisons support statements about the specified equilibrium and property calculations at their measured and searched states. They combine calibration, fixed-parameter prediction of observations outside the objective, comparison-only properties and numerical phase verification. Previously accessed held-out results do not become blind validation merely because their residuals are absent from the objective. Finite phase searches do not establish a continuous phase boundary, and the PZ counterexample shows that the search may miss a resolved split. The results do not qualify absorber or stripper design, kinetic or transport coefficients, solvent blends, degradation or solid equilibrium. These limits define the demonstrated transfer.

5. Conclusions

A frozen description of water, CO_2 and inorganic ions can accompany amine-specific fitting across MDEA, AMP, PZ and DEA, but calibration and transfer remain property- and source-dependent. Residual loading-shape errors persist in the fitted pressures, and PZ species residuals remain large relative to their assumed measurement scales. Shared-parameter substitutions followed by refitting demonstrate compensation: an apparently improved unfitted AMP score before refitting does not survive restoration of its calibration fit. For MDEA, which tests the shared inorganic-carbon description without an amine-carbamate pathway, substituting Part I's bound bicarbonate–water interaction and R2 law moves the unfitted error in the direction opposite to AMP before refitting, and the MDEA cation–bicarbonate interaction reaches its bound; this pattern does not identify a uniquely defective bicarbonate parameter.

MDEA's held-out pressure AARDs are 28.19 % for Xiao and 18.27 % for Ermatchkov, with the latter mixing measured-range and extrapolated dissociation conditions. The comparison-only Kamps PZ total pressures give 42.5 % AARD and −40.3 % bias, including substantial underprediction inside both PZ law ranges. AMP's held-out Kundu error is 42.85 %, while PZ gives 21.94 % against held-out Ermatchkov pressures and 48.52 % at 393.15 K. DEA gives 79.54 and 44.17 % on the held-out Seo and Hong and Lal data, despite 0.98 % fitted integral-heat AARD, and predicts the wrong sign of water–DEA excess enthalpy.

Local conditional parameter bands are much smaller than the AMP and DEA unfitted discrepancies, overlap the MDEA errors, and are limited by large PZ sensitivities. At least forty false liquid splits are detected for the reported PZ fit, including thirty-six high-loading Kamps states; compression to 1.5 times bubble pressure preserves all four Hilliard splits, and direct evaluations demonstrate that the standard finite search misses two of them. No additional phase was detected in the MDEA, AMP and DEA searches. The extent of transfer is therefore defined by the reported calibration errors, unfitted property failures, reaction-law extrapolations and conditional uncertainty, rather than by a universal accuracy or process-use claim.

Data and code availability

Processed observations, parameter files, numerical results and presentation scripts are provided with the public electronic data at `https://github.com/tannerpolley/Amine-Thermodynamics`. Presentation code reads the stored calculations without evaluating the thermodynamic model again. Input checksums and the source of every displayed numerical result are listed in the accompanying numerical-source index. The compiled `epcsaft` library evaluates the equation of state and its equilibrium derivatives; Supplementary Section S8 records the immutable file identity used in the calculations. Deposit identifier: *[Zenodo DOI supplied in the final submission]*. Referee access to the compiled ePC-SAFT library: *[access arrangement supplied in the final submission]*. Third-party measurements retain their source terms.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT author statement

Tanner W. Polley: Conceptualization, methodology, software, formal analysis, investigation, data curation, writing–original draft, writing–review and editing, visualization.

Declaration of generative AI and AI-assisted technologies

During preparation of this manuscript, OpenAI Codex and Claude (Anthropic) were used to draft and revise manuscript text, check computational consistency, and assist with source checking. The author reviewed and edited the content and takes responsibility for the final manuscript.

Nomenclature

α: added CO_2 per total amine, mol/mol; w_A: unloaded amine mass fraction; x_i, y_i: liquid and vapor true-species mole fractions; P: total pressure; $p_{CO_2} = y_{CO_2} P$: partial pressure; K: equilibrium constant on the stated standard state; C: least-squares cost; s: assigned observation scale; SE: conditional standard error; AARD: average absolute relative deviation; NMR: nuclear magnetic resonance; k_{ij}: unlike-pair dispersion correction; d^{Born}: Born diameter.

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